Aritra Das

Ph.D. Candidate
Department of Physics
University of Maryland, College Park

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Education

University of Maryland — **College Park, MD, USA**
Ph.D. Physics — August 2022–present

Advisor: Dr. Victor V. Albert
Anticipated graduation : 06/2027

Indian Institute of Science — **Bengaluru, KA, India**
BS (RESEARCH), Major in Physics — Aug 2018–July 2022

Thesis advisor: Dr. Aninda Sinha

Research Interests

I am interested in the ***physics of learning***, ***machine learning***, and the ***application of AI to physics and mathematics***, with an emphasis on both theoretical foundations and practical systems.

My research spans **influence of data on learning**. This includes *representation learning*, *emergent abilities*, *neural network architectures*, *mechanistic interpretability*, *interpretable AI*, *geometric deep learning*, *group theory*, *finite-field modular arithmetic*, *homogeneous spaces*, and *manifold learning*.

I am also interested in *AI for scientific discovery*, particularly *quantum computing*, and *Quantum Random Circuit Sampling (QRCS)*.

Publications

UNDER REVIEW

AD, Francesco Cagnetta, Darshil Doshi, Victor V. Albert, Maissam Barkeshli, Andrey Gromov. What's in a Tree? Hierarchical n-gram models as controlled approximators of language. Under review at ICLR 2027.

AD, Joseph T. Iosue, Victor V. Albert. HomID : Benchmarking Intrinsic Dimension Estimators on Homogenous Manifolds with Anisotropic Embeddings. Under review at TMLR. https://arxiv.org/abs/2510.01335v2

PUBLISHED

AD, Francesco Cagnetta, Darshil Doshi, Victor V. Albert, Maissam Barkeshli, Andrey Gromov. Treegrams: Hierarchical n-gram models as controlled approximators of language. Workshop on Lingustic Principles of Foundation Models at Neurips 2026. https://openreview.net/forum?id=pPG1iq5N08&noteId=nP5k6gAvnh

Tianyu H., Darshil D., **AD**, Andrey G. Learning to grok: Emergence of in-context learning and skill composition in modular arithmetic tasks, NeurIPS 2024 (Oral). https://arxiv.org/abs/2406.02550

Tianyu H., Darshil D., **AD**, Andrey G. Exploring model depth and data complexity through the lens of cellular automata, NeurIPS 2024 Workshop. https://openreview.net/forum?id=SGoI97b5KK

Darshil D., Tianyu H., **AD**, Andrey G. Grokking Modular Polynomials, ICLR 2024 Workshop. https://arxiv.org/abs/2406.03495

Darshil D., **AD**, Tianyu H., Andrey G. To Grok or not to Grok: Disentangling Generalization and Memorization on Corrupted Algorithmic Datasets, ICLR 2024. https://arxiv.org/abs/2310.13061

AD, Umberto B., Sergej M. Fractionalized holes in one-dimensional $\mathbb{Z}_2$ gauge theory coupled to fermionic matter: Deconfined dynamics and emergent integrability, Phys. Rev. B 107, 064302. https://journals.aps.org/prb/abstract/10.1103/PhysRevB.107.064302

Work in Progress

Quantum-enhanced random circuit sampling with machine learning.

Mathematical data for uncovering structured representations in AI.

Technical Skills

Programming	Python, PyTorch, NumPy, SciPy, scikit-learn, Git, Bash, LaTeX
Machine Learning	Deep learning, Transformers, Mechanistic Interpretability, Grokking, In-context learning, Generalization and memorization, Model evaluation
Scientific Computing	High-dimensional data analysis, Intrinsic dimension estimation, manifold learning, numerical experiments, SLURM/HPC workflows
Mathematical Background	Quantum information, Statistical physics, Geometry, Topology, Probability, Linear Algebra

Memberships

2025- Simons Collaboration on Physics of Learning and Neural Computation

Awards & Fellowships

2024 Kapo-Barwick Endowment Award, Department of Physics, University of Maryland
2022 Ralph Myers and Friends of Physics Outstanding Teaching Assistant, Department of Physics, University of Maryland
2021 DAAD WISE Virtuell Scholar, German Academic Exchange Service (DAAD)

Teaching Experience

Fall 2026	Electromagnetism, *Teaching Assistant*
Spring 2025	Quantum and Statistical Physics II, *Teaching Assistant*
Spring 2025	Fundamentals of Physics I, *Teaching Assistant*
Fall 2023	Physics of Music Laboratory, *Teaching Assistant*
Summer 2023	Fundamentals of Physics I, *Teaching Assistant*
Spring 2023	Fundamentals of Physics for Life Sciences I, *Teaching Assistant*
Fall 2022	General Physics: Electrodynamics, Light, Relativity and Modern Physics, *Teaching Assistant*

Outreach & Professional Development

PROFESSIONAL SERVICES

Reviewer Neurips, ICML, ICLR. Awarded "Silver Reviewer" for ICML 2026.

CONFERENCES AND WORKSHOPS

September 2026	Simons' Foundation Annual Meeting for the Simons Collaboration on Physics of Learning and Neural Computation, New York City, NY
August 2026, June 2023	Princeton Machine Learning Theory Summer School, Princeton, NJ
July 2026	Simons Collaboration on Physics of Learning and Neural Computation Workshop, National Taiwan University, Taipei, Taiwan
July 2026	International Conference of Machine Learning, Seoul, South Korea
November 2025	Simons Collaboration on Physics of Learning and Neural Computation Core Workshop, Stanford, USA
December 2024	Neural Information and Processing Systems, Vancouver, Canada
May 2024	International Conference on Learning Representations, Vienna, Austria
August 2023	IAIFI Summer School and Workshop, Boston, MA

SERVICE AND OUTREACH

February 2025	Graduate Student Colloquium Committee, *Organizer*
January 2023	GRAD-MAP Winter Workshop, *Mentor*
2020–21	Quarks, *Editorial Member*
2019–20	Pravega, *Organizational Member*

References

Victor V. Albert
Adjunct Assistant Professor, Fellow
QuICS, NIST and UMD
valbert4@gmail.com

Andrey Gromov
Research Scientist
META Super Intelligence Lab
gromovand@gmail.com

Maissam Barkeshli
Research Scientist
Anthropic
mbarkeshli@gmail.com